THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a right triangle with hypotenuse 39 cm and one leg 15 cm (in cm²) is: 1

(a) 540 **(b)** 292 **(c)** 270 **(d)** 285

2. Which set of numbers is a Pythagorean triplet? 1

(a) 36, 84, 91 **(b)** 35, 86, 91 **(c)** 35, 84, 92 **(d)** 35, 84, 91

3. An isosceles triangle has equal sides 40 cm and base 48 cm. Its height (in cm) is: 1

(a) 33 **(b)** 32 **(c)** 31 **(d)** 16

4. A rectangle has a diagonal of 40 cm and one side 24 cm. Its perimeter (in cm) is: 1

(a) 56 **(b)** 128 **(c)** 112 **(d)** 114

5. The side of a rhombus is 40 cm and one diagonal is 48 cm. The other diagonal (in cm) is: 1

(a) 64 **(b)** 65 **(c)** 66 **(d)** 32

6. Two vertical poles of heights 38 m and 14 m stand 32 m apart on level ground. The distance between their tops (in m) is: 1

(a) 41 **(b)** 56 **(c)** 38 **(d)** 40

7. The area of a right triangle with hypotenuse 40 cm and one leg 24 cm (in cm²) is: 1

(a) 384 **(b)** 408 **(c)** 480 **(d)** 768

8. Which set of numbers is a Pythagorean triplet? 1

(a) 40, 75, 85 **(b)** 41, 75, 85 **(c)** 40, 77, 85 **(d)** 40, 75, 86

9. An isosceles triangle has equal sides 35 cm and base 42 cm. Its height (in cm) is: 1

(a) 14 **(b)** 29 **(c)** 27 **(d)** 28

10. A rectangle has a diagonal of 35 cm and one side 21 cm. Its perimeter (in cm) is: 1

(a) 98 **(b)** 112 **(c)** 49 **(d)** 100

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The diagonal of a rectangle is 35 cm and one side is 21 cm. Find the other side and the area. 2

12. The side of a rhombus is 50 cm and one diagonal is 28 cm. Find the other diagonal. 2

13. An isosceles triangle has equal sides of 40 cm and base 48 cm. Find its height. 2

14. The diagonal of a rectangle is 50 cm and one side is 30 cm. Find the other side and the area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 13 m long reaches a point 12 m up a wall. The foot is then pulled away so that it is 12 m from the wall. How far down does the top of the ladder slip? 3

16. Show that the triangle with sides 24 cm, 32 cm and 40 cm is right-angled and find its area. 3

17. The altitude drawn to the hypotenuse of a right triangle with legs 24 cm and 32 cm: find the hypotenuse and the length of this altitude. 3

OR

A ladder 17 m long reaches a point 15 m up a wall. The foot is then pulled away so that it is 15 m from the wall. How far down does the top of the ladder slip? 3

18. Show that the triangle with sides 21 cm, 28 cm and 35 cm is right-angled and find its area. 3

OR

The altitude drawn to the hypotenuse of a right triangle with legs 21 cm and 28 cm: find the hypotenuse and the length of this altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A man walks 24 m east and then 32 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 50 cm and base 60 cm. Find its height, area and perimeter. 5

20. A man walks 21 m east and then 28 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 50 cm and base 28 cm. Find its height, area and perimeter. 5

END OF QUESTION PAPER — Total: 40 Marks

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